

Echo Detection Using the Herglotz Wavefunction in Spatial Room Impulse Responses Measured with Spherical Microphone Arrays

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Abstract—Early reflections in spatial room impulse responses (SRIRs) measured using spherical microphone arrays (SMAs) play an important role in spatial audio analysis, rendering, and reverberation modeling. Their accurate localization and characterization can facilitate the analysis, processing, and manipulation of measured reverberation fields. This paper proposes a method for detecting and characterizing early reflections based on the Herglotz wavefunction formalism. The measured sound field is represented as a continuous superposition of incident plane waves, from which a localization function is derived in the spherical harmonic domain. An adaptive radial Gaussian fitting procedure is then used to estimate both the directions of arrival (DoAs) and the number of incident reflections within a single analysis frame. The proposed framework is evaluated using simulated and measured SRIRs and compared with conventional steered response power (SRP) and multiple signal classification (MUSIC) localization methods. The results demonstrate improved localization accuracy and more reliable detection of multiple simultaneous reflections, highlighting the potential of the proposed Herglotz-based formulation for the analysis of early reflections in SMA-measured SRIRs.

Index Terms—spatial room impulse responses, spherical microphone arrays, higher-order Ambisonics, Herglotz wavefunction, echo detection, direction-of-arrival estimation, spatial audio, room acoustics.

I. INTRODUCTION

THIS paper addresses the detection and characterization of early reflections in spatial room impulse responses (SRIR) measured using spherical microphone arrays (SMA). Specifically, it introduces an approach based on the Herglotz wavefunction formalism, which represents the sound field as a superposition of plane waves equipped with a kernel function whose values specify the strength of the incident field for each direction on the sphere.

This work forms part of a broader research effort aimed at developing a unified space-time-frequency framework for the analysis, processing, and manipulation of SRIRs. The objective of this framework is to provide detailed analysis and modeling tools of SMA-measured SRIRs, thereby

facilitating the reproduction of spatial reverberation effects. Such reproduction may be achieved either through direct multichannel convolution with measured SRIRs, or through hybrid approaches in which the late reverberation tail is synthesized by a feedback delay network (FDN) [1].

In direct convolution-based reproduction, the non-decaying measurement noise floor must be removed or compensated for, e.g. through late-reverberation resynthesis techniques [2]–[4]. Beyond noise mitigation, these techniques can also be employed to modify the spatio-temporal and spectral characteristics of the late reverberation tail. Hybrid reverberators offer additional advantages, including reduced computational complexity and real-time control over reverberation decay characteristics through the manipulation of FDN parameters.

In both cases, a detailed “cartography” of the dominant early reflections significantly enhances the capabilities of the framework. Such information has considerable potential for supporting operations such as reflection filtering, redistribution, and editing, while also enabling the definition of higher-level objective or perceptual descriptors for controlling these transformations.

Although the long-term objective is to construct complete early-reflection cartographies through frame-by-frame analysis of the early portion of the SRIR, the present work focuses on the first fundamental step: localizing incident reflections within a single analysis frame. Here, *localization* primarily refers to the estimation of the reflections’ directions of arrival (DoAs), whereas *characterization* encompasses the set of attributes used to describe an individual reflection, namely its DoA, relative time of arrival (ToA), total energy, and spectral envelope.

A. Related Work

The problem addressed in this paper belongs to the general class of source localization problems, for which an extensive body of literature exists covering a wide range of microphone array configurations and signal processing techniques. However, several characteristics unique to echo detection in SMA-measured SRIRs restrict the set of applicable methods in practice.

These limitations are discussed in further detail in Secs. II and III. They are primarily related to the fact that early reflections in an SRIR are both impulsive, with very limited temporal support, and highly correlated with

the direct sound, and consequently with each other. In contrast, many state-of-the-art localization techniques rely on the estimation of a spatial covariance matrix over extended time windows and assume statistical independence between sources. Furthermore, several approaches do not explicitly address the estimation of the number of active sources, which is a critical requirement for blind analysis of SRIRs.

Two widely used parametric approaches for SRIR analysis and reproduction are spatial impulse response rendering (SIRR) [5] and the spatial decomposition method (SDM) [6]. Both techniques typically estimate at most one dominant DoA per analysis frame, although higher-order SIRR (HO-SIRR) [7] extends this framework to allow multiple simultaneous DoAs. SIRR and HO-SIRR rely on the estimation of a pseudo-intensity vector (PIV), which in term requires relatively long analysis windows. Similarly, McCormack et al. [8] proposed an “angular sharpening” approach for steered response power (SRP) localization maps, which also depends on spherical sector-based PIV estimation, in a manner analogous to HO-SIRR.

Subspace methods such as multiple signal classification (MUSIC) [9] and the estimation of signal parameters via rotational invariance techniques (ESPRIT) [10] enable the localization of multiple simultaneous sources. However, these methods also require covariance matrix estimation, which again presupposes sufficiently long analysis windows. In addition, MUSIC- and ESPRIT-based methods generally assume that the sources are either uncorrelated or, at most, weakly correlated.

Nevertheless, several adaptations of MUSIC-inspired methods have been proposed for SRIR analysis. Khaykin and Rafaely [11] introduced a frequency-smoothing approach to estimate the covariance matrix from a single-snapshot full-length SRIR. However, due to the frequency-dependent characteristics of the spherical array response in SMA measurements (see Sec. II-B), the full frequency bandwidth cannot be fully exploited. Furthermore, the maximum number of detectable reflections remains fundamentally limited by the maximum spherical harmonic (SH) order of the SMA in this formulation.

To address these limitations, Huleihel and Rafaely [12] proposed a windowed time-smoothing strategy to construct a MUSIC pseudospectrum. Morgenstern and Rafaely [13] later introduced a modal smoothing technique in the SH (or wave spectral) domain requiring a spherical loudspeaker array (SLA) and therefore restricting its applicability to a multiple-input multiple-output (MIMO) setting [14]. Notably, neither approach provides a complete framework for estimating both the number of reflections and their associated DoAs.

More recently, Shlomo and Rafaely proposed a blind localization method for early room reflections based on phase-aligned spatial correlation [15]. Their approach demonstrates the feasibility of estimating multiple reflection directions from SMA-measured SRIRs without requiring prior knowledge of the reflection structure. However, it relies on phase alignment and spatial correlation of the

measured signals, whereas the present work formulates the localization problem directly in the SH domain through a Herglotz wavefunction representation.

B. Overview of the Proposed Approach

Motivated by the limitations of existing approaches, this paper introduces a localization framework based on the Herglotz wavefunction formalism for detecting and characterizing multiple early reflections within a single SRIR analysis frame. The Herglotz representation describes the incident sound field as a continuous angular distribution of plane waves, allowing the localization problem to be formulated as the estimation of a directional function on the sphere rather than as the direct identification of a fixed number of discrete sources.

The proposed framework consists of three main steps. First, the measured SMA sound field is transformed into a Herglotz-based localization function in the spherical harmonic domain. Second, peaks in the resulting localization function are detected using a sparse set of spherical Gaussian components, whose parameters provide estimates of the directions and relative strengths of incident reflections. Finally, an adaptive information-criterion-based procedure estimates the number of Gaussian components and, consequently, the number of reflections present in the analysis frame. In contrast to SRP-based analysis, which evaluates the sphere by considering one direction independently at a time, and can therefore suffer from smearing and spurious peaks when highly correlated echoes overlap, the Herglotz approach formulates localization as a global inverse problem. We therefore solve the global inverse problem for the optimal continuous directional distribution over the entire sphere simultaneously, enabling more accurate resolution of multiple concurrent reflections.

Although the long-term objective is to construct complete early-reflection cartographies through frame-by-frame analysis of SRIRs, the present work focuses on the more fundamental problem of localizing and characterizing multiple reflections occurring within a single analysis frame.

The paper is structured as follows. Section II introduces the theoretical background underlying the proposed method. After a description of SRIR temporal segmentation in Section II-A, the sound field generated by the scattering of a plane wave by a rigid sphere is reviewed. Section III-A introduces the Herglotz representation of sound fields. These formulations are combined in Section III, where the proposed localization framework and echo detection procedure are developed.

In Section III-A, the field generated by multiple independent plane waves is expressed in the spherical harmonic (SH) domain. Section III-B extends this formulation to the more general case of echo-driven sound fields, where reflections are modeled as a set of localized contributions that may be represented using radial Gaussian basis functions on the sphere. Importantly, the number of echoes is not assumed to be known a priori. Sections III-C and

III-D present the resulting detection procedures and the associated adaptive estimation of the number of echoes. Finally, Section IV presents an evaluation of the proposed method.

II. THEORETICAL BACKGROUND

A. SRIR Time Segmentation

Throughout this work, the SRIR is assumed to be temporally segmented into an *early* region, consisting predominantly of the direct sound and its first discrete reflections, followed by a stochastic, exponentially decaying late reverberation tail, as originally described by Sabine, Eyring, and later formalized by Schroeder [16]. The transition between these two regimes is commonly referred to as the *mixing time*, t_{mix} , following the interpretation introduced by Polack [17].

The mixing time considered here is defined from the temporal evolution of a measure of the SRIR's spatial incoherence, following the approaches of Götz et al. [18] and Massé et al. [2], [3]. It is measured relative to ToA of the direct sound, t_0 , and is therefore specific to each SRIR.

A complete analysis of the early reflections consequently requires frame-by-frame characterization over the interval $t \in [t_0, t_{\text{mix}}]$. Such an analysis involves several additional processing steps that are beyond the scope of the present work, including the selection of analysis frames, the rejection of false detections, and the association of repeated detections corresponding to the same physical reflection. Some of these aspects have previously been discussed by Massé [4].

B. Plane-Wave Field Measurement with an SMA

1) *SH representation of a single plane wave*: The total sound field at an arbitrary point $\mathbf{r} = (r, \boldsymbol{\Omega}) = (r, \theta, \phi)$ outside a *rigid* sphere with radius r_s resulting from a single unit amplitude plane wave incident from direction $\mathbf{d} = (1, \hat{\boldsymbol{\Omega}}) = (1, \theta_d, \phi_d)$ is obtained as the sum of incident and scattered contributions:

$$\begin{aligned} u(kr, \mathbf{r}, \mathbf{d}) &= u_{\text{in}}(kr, \mathbf{r}, \mathbf{d}) + u_{\text{sc}}(kr, \mathbf{r}, \mathbf{d}) \\ &= \sum_{\ell=0}^{\infty} b_{\ell}(kr) \sum_{m=-\ell}^{\ell} Y_{\ell m}(\boldsymbol{\Omega}) Y_{\ell m}^*(\hat{\boldsymbol{\Omega}}), \end{aligned} \quad (1)$$

where

$$b_{\ell}(kr) = 4\pi i^{\ell} \left[j_{\ell}(kr) - \frac{j'_{\ell}(kr_s)}{h'_{\ell}(kr_s)} h_{\ell}(kr) \right] \quad (2)$$

is the so-called mode strength (or holographic function) for a rigid sphere [19, Eq. (6.186)]. The j_{ℓ} and h_{ℓ} denote the spherical Bessel and Hankel functions of order ℓ , respectively, and the prime denotes differentiation of the corresponding function with respect to its argument.

On the other hand, for a function u defined on the unit sphere, the coefficients $u_{\ell m}$ can be obtained from the orthonormality of the spherical harmonics:

$$u_{\ell m} = \int_{\mathbb{S}^2} u(\boldsymbol{\Omega}) Y_{\ell m}^*(\boldsymbol{\Omega}) d\boldsymbol{\Omega}.$$

In practice, the sound field u is sampled by the Q transducers of the SMA located at positions $\mathbf{r}_q$ on the sphere with radius $r = r_s$. The theory of optimal sampling on the sphere is beyond the scope of this work; relevant references include Meyer and Elko [20], Abhayapala [21], Rafaely [22], [23], and Chardon et al. [24].

For practical applications, the sum over ℓ in Eq. (1) must be truncated at some maximum order L . The maximum attainable SH order is determined by factors such as the accuracy of the quadrature scheme and the spatial sampling density; see, e.g., [24]. For example, the 32-capsule Eigenmike (EM32) supports accurate SH representations up to $L = 4$, whereas the 64-capsule Eigenmike (EM64) is practically limited to approximately $L \approx 6$ due to the poorer conditioning of its encoding matrix. As a simple rule of thumb for determining the required SH order, we refer to [25, Eq. (17)].

2) *Compensation of sphere radius*: Importantly, only the mode strength b_{ℓ} depends on the sphere radius. Consequently, if the SH coefficients are compensated by an inverse model of b_{ℓ} , the resulting representation becomes independent of the specific SMA geometry [26]. In practice, this compensation is achieved by applying encoding (or correction) filters [20], [27] after the SH transform. Denoting the frequency response of these filters by $a_{\ell}(f)$, the effective modal response becomes $c_{\ell}(f) = a_{\ell}(f)b_{\ell}(f)$, yielding the corrected SH coefficients $\tilde{u}_{\ell m}(f, \hat{\boldsymbol{\Omega}})$. The filters are typically designed such that $c_{\ell}(f)$ is as flat and as geometry-independent as possible over the usable frequency range.

In practice, the frequency response $c_{\ell}(f)$ is not perfectly flat, resulting in a frequency-dependent function (see, for example, Daniel and Moreau [27], among many others):

$$w(f, \boldsymbol{\Omega}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_{\ell}(f) Y_{\ell m}(\boldsymbol{\Omega}) Y_{\ell m}^*(\hat{\boldsymbol{\Omega}}). \quad (3)$$

Fig. 1 shows $w(f, \Theta)$ for an $L = 4$ HOA-encoded plane wave, evaluated along the axisymmetric great circle defined with respect to the reference direction $\hat{\boldsymbol{\Omega}}$. The directivity pattern is simulated for an mh-acoustics EM32, and the corresponding directivity index (DI) [28] is shown in Fig. 2. Based on the DI, a “maximally directive” frequency band can be identified. The upper bound is given by the spatial aliasing frequency f_{alias} [29], beyond which the directional response becomes increasingly ambiguous. A lower bound f_{minMaxDI} is then defined such that the DI remains at least as high as its value at the aliasing limit, i.e., $\text{DI}(f_b) \geq \text{DI}(f_{\text{alias}})$ for all $f \in [f_{\text{minMaxDI}}, f_{\text{alias}}]$.

III. PROPOSED METHOD

A. Herglotz Wavefunction

As discussed in Section II-A, the predominant early reflections in an SRIR are assumed to be well approximated by ideal impulsive arrivals. Consequently, within a single analysis frame, the sound field measured by an SMA can be interpreted as a superposition of multiple plane waves arriving from different directions, each contributing over

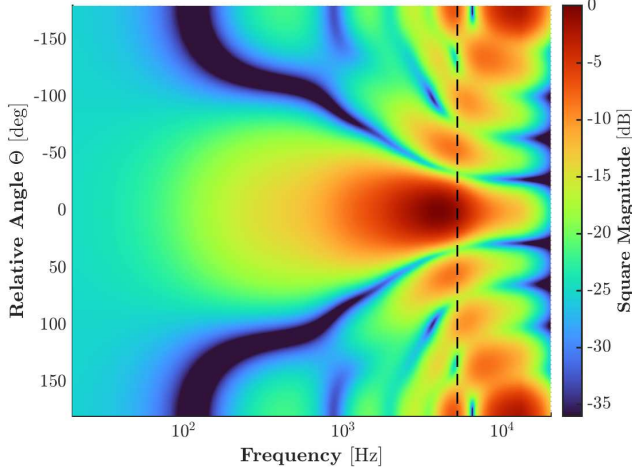


Fig. 1. $w(f, \Theta)$ (Eq. 3) evaluated along the axisymmetric great circle Θ with respect to the incident direction $\hat{\Omega}$, for a simulated $L = 4$ Higher-Order Ambisonics (HOA) encoding [27] of a plane wave measured with the mh-acoustics EM32 spherical microphone array (SMA). The dashed line indicates the spatial aliasing frequency, f_{alias} .

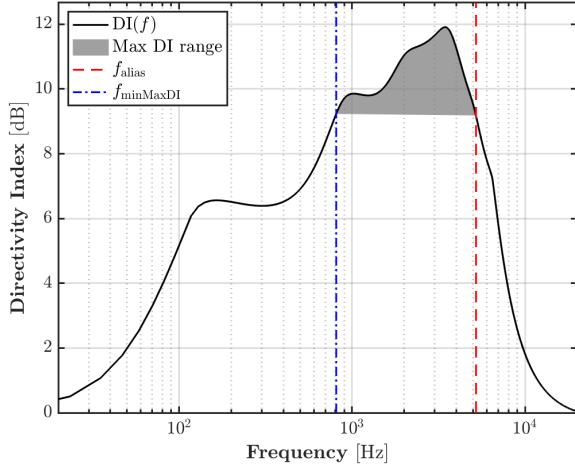


Fig. 2. Directivity index (DI) as a function of frequency for the simulated order $L = 4$ HOA encoding of a plane wave as measured by the mh-acoustics EM32 SMA. The red dashed line indicates the spatial aliasing frequency f_{alias} , the blue dash-dotted line the lower bound f_{minMaxDI} of the SMA's operational range. The shaded region denotes the frequency band of maximal directivity, such that $\text{DI}(f_b) \geq \text{DI}(f_{\text{alias}}) \forall f_b \in [f_{\text{minMaxDI}}, f_{\text{alias}}]$.

the analyzed frequency range. This naturally motivates the use of the Herglotz wavefunction formalism [30, p. 60]. The Herglotz wavefunction is defined as

$$u(k, \mathbf{r}) = \int_{\mathbb{S}^2} g(\mathbf{d}) e^{i\mathbf{k}\mathbf{r}\cdot\mathbf{d}} d\mathbb{S}^2 \quad (4)$$

where the sound field u at position $\mathbf{r}$ is represented as a continuous superposition of plane waves arriving from all directions $\mathbf{d} \in \mathbb{S}^2$. The direction-dependent weighting function $g(\mathbf{d})$, referred to as the Herglotz kernel, describes the contribution of each plane-wave component as a con-

tinuous function. The Herglotz kernel is assumed to satisfy $g \in L^2(\mathbb{S}^2)$, i.e., $g(\mathbf{d})$ must be square integrable over the unit sphere.

In [30, Lemma, 3.20], it is shown that if an incident sound field is a Herglotz wavefunction with kernel $g(\mathbf{d})$, then the field scattered by a sphere, and hence the resulting total field, can also be expressed as a Herglotz wavefunction with the same kernel

$$u(k, \mathbf{r}) = \int_{\mathbb{S}^2} g(\mathbf{d}) u^{\text{pw}}(\mathbf{r}, \mathbf{d}) d\mathbb{S}^2(\mathbf{d}), \quad (5)$$

where $u^{\text{pw}}(\mathbf{r}, \mathbf{d})$ denotes the acoustic field at $\mathbf{r}$ resulting from the scattering of a single plane wave arriving from direction $\mathbf{d}$.

This formulation also holds for the sound field evaluated on the sphere. Thus, given measurements of the sound field over the SMA surface, the problem of finding directions of the incident reflections can be formulated as an inverse problem in which the Herglotz kernel is estimated from the measurements. The dominant DoAs can then be identified from the principal maxima of the Herglotz kernel g .

Applying the spherical harmonic transform (SHT) to both sides of Eq. (5) yields

$$u_{\ell m}(f) = \int_{\mathbb{S}^2} g(\hat{\Omega}) u_{\ell m}^{\text{pw}}(kr_s, \hat{\Omega}) d\hat{\Omega}. \quad (6)$$

Here, the incident direction $\mathbf{d}$ has been replaced by $\hat{\Omega}$ to emphasize that the integration is performed over all possible plane-wave directions.

The estimation of the Herglotz kernel $g(\hat{\Omega})$ requires a numerical discretization of the integral over the unit sphere. This differs subtly from conventional beamforming methods, which typically evaluate the sound field over a fixed grid of scanning directions. In the Herglotz formulation, the discretization of the unit sphere is determined solely by the accuracy required for numerical quadrature.

Furthermore, since the proposed method operates on HOA-encoded SRIRs, the SH coefficients are assumed to have been compensated by the encoding filters described in Section II-B. Consequently, the mode strengths $b_{\ell}(f)$ are replaced by the effective frequency responses $c_{\ell}(f)$. This compensation is indicated in the following by using $\tilde{u}_{\ell m}$ instead of $u_{\ell m}$ to denote the spherical harmonics coefficients.

The sampled kernel provides a directional interpolation of the incident field, where a single echo corresponds to a localized contribution in this domain. The relationship between the measured SH coefficients and the sampled kernel can be written in matrix form as

$$\tilde{\mathbf{u}}(f) = \mathbf{D}(f) \mathbf{g}, \quad (7)$$

where the (n, j) -th entry of the dictionary matrix $\mathbf{D}(f)$ is given by

$$\mathbf{D}_{nj} = \beta_j c_{\ell}(f) Y_{\ell m}^*(\hat{\Omega}_j).$$

The multi-index

$$n = \ell^2 + \ell + m + 1, \quad \ell = 0, \dots, L, \quad m = -\ell, \dots, \ell$$

enumerates all spherical harmonics up to order L (cf. [21], [31]). The quantities β_j and $\hat{\Omega}_j$ denote the D quadrature weights and nodes associated with the discretization of the integral over the sphere in Eqs. (5) and (6), respectively. The vector

$$\tilde{\mathbf{u}}(f) = [\tilde{u}_{00}(f), \dots, \tilde{u}_{LL}(f)]^\top$$

contains the modified spherical harmonic coefficients, while

$$\mathbf{g} = [g(\hat{\Omega}_1), \dots, g(\hat{\Omega}_D)]^\top.$$

contains the sampled values of the Herglotz kernel.

B. Echo Localization Function

If the early reflections were simply modeled as a superposition of independent plane waves, it would be sufficient to determine $\mathbf{g}$ based on Eq. (7). However, the early segment of an SRIR generally consists of multiple reflections that overlap within a given analysis frame and cannot necessarily be interpreted as independent plane-wave components. We therefore assume that each analysis frame in the early segment, $t \in [t_0, t_{\text{mix}}]$, contains P impulsive echoes, where each echo can be represented by a Herglotz wavefunction with an associated kernel that is invariant with respect to frequency. Each echo is characterized by a frequency-dependent amplitude $a_p(f)$ and a discretized Herglotz kernel

$$\mathbf{g}_p = [g_p(\hat{\Omega}_1), \dots, g_p(\hat{\Omega}_D)]^\top, \quad p = 1, \dots, P.$$

The resulting SH coefficients of the encoded sound field are therefore given by

$$\tilde{u}_{\ell m}(f) = c_\ell(f) \sum_{j=1}^D \beta_j Y_{\ell m}^*(\hat{\Omega}_j) \sum_{p=1}^P a_p(f) g_p(\hat{\Omega}_j). \quad (8)$$

Equivalently, this expression can be written in matrix form as

$$\begin{aligned} \tilde{\mathbf{u}}(f) &= \mathbf{D}(f) \mathbf{G} \mathbf{a}(f) \\ &= \mathbf{D}(f) \mathbf{m}(f), \end{aligned} \quad (9)$$

where $\mathbf{G}_{ij} = \mathbf{g}_j(\hat{\Omega}_i)$ contains the discretized kernels of the individual echoes, $\mathbf{a}(f)$ contains the frequency-dependent amplitudes $a_p(f)$, and $\mathbf{m}(f) = \mathbf{G} \mathbf{a}(f)$ is a vector of length D that will henceforth be referred to as the *localization function*. This function combines the directional contributions of all echoes and therefore contains the information required to identify their DOAs. Importantly, the number of echoes P is only implicitly "encoded" in $\mathbf{m}$ and need not to be known a priori; instead, it is estimated adaptively as described in Section III-D.

The first step of the proposed method is to estimate the localization function $\mathbf{m}(f)$ from the SH coefficients measured by the SMA. The dictionary matrix $\mathbf{D}(f)$ has $(L+1)^2$ rows and D columns, where D denotes the number of quadrature nodes on the sphere used to interpolate the Herglotz kernel.

We estimate the localization function using Tikhonov regularization [32]:

$$\mathbf{m}(f) = (\mathbf{D}^H \mathbf{D} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{D}^H \tilde{\mathbf{u}}(f), \quad (10)$$

where $\mathbf{T} = \mu \mathbf{I}^{D \times D}$ is the regularization operator, μ is the regularization parameter, and $\mathbf{I}^{D \times D}$ the $D \times D$ identity matrix. This formulation favors solutions with reduced norm and stabilizes the inversion of the ill-conditioned system.

In this formulation, the regularization operator effectively acts as a spatial smoothing constraint on the estimated localization function. This is consistent with the smoothness properties of the Herglotz kernel and implies that, although a finer sampling scheme may be used to represent the solution, the achievable spatial resolution remains fundamentally limited by the maximum SH order.

Finally, the frequency dependence has so far been treated continuously for notational simplicity. In practice, frame-based analysis (e.g., using a short-time Fourier transform, STFT) discretizes the frequency axis into K bins f_k over the positive-frequency range. The localization function could therefore be estimated independently for each frequency bin using Eq. (10). However, as discussed in Section II-B, the directivity of an HOA-encoded plane wave depends strongly on frequency, and directional information becomes unreliable above the spatial aliasing frequency f_{alias} . Consequently, estimation of $\mathbf{m}(f_k)$ is restricted to the maximally directive frequency band defined in Section II-B and illustrated in Fig. 2, i.e. for $f_k \in [f_{\text{minMaxDI}}, f_{\text{alias}}]$ such that $\text{DI}(f_k) \geq \text{DI}(f_{\text{alias}})$.

C. Echo Detection

The next step consists of identifying the directional maxima of the localization function $\mathbf{m}(f_k)$ in order to estimate the DOAs of the incident echoes present within the current analysis frame. Since the magnitude of each localized contribution is related to the corresponding echo strength, this procedure also provides a first characterization of their relative amplitudes.

In general, the localization function $\mathbf{m}(f_k)$ is complex-valued, since the HOA-encoded response $c_\ell(f)$, the SH basis functions, and the STFT coefficients $\tilde{\mathbf{u}}(f)$ are complex valued quantities. However, the directional information indicating the presence of an echo is contained in the magnitude of the localization function rather than its phase. Furthermore, recall that the Herglotz kernels $g_p(\hat{\Omega})$ associated with the individual echoes are assumed to be frequency invariant. Consequently, the detection procedure should combine information across frequency while remaining independent of the particular frequency bin. We define a frequency-averaged localization function as

$$\bar{m}_d = \frac{1}{\Gamma_{\text{tot}}} \sum_k \gamma_k |m_d(f_k)|, \quad (11)$$

where the summation is performed over the maximal directive frequency range $f_k \in [f_{\min}, f_{\max}]$, and where

$$\gamma_k = \max_d (|m_d(f_k)|), \quad \Gamma_{\text{tot}} = \sum_k \gamma_k.$$

The weighting factor γ_k emphasizes frequency bins with stronger directional information, while reducing the influence of bins where the localization function is dominated by noise or insufficient directivity. The index d denotes the d -th sampled direction of the localization grid.

Having defined the frequency-independent localization function $\bar{\mathbf{m}}$, we now formulate the detection of its directional peaks as a parametric estimation problem. We model the localization function as a sum of P spherical radial basis functions (RBFs), using spherical Gaussian functions to represent the contribution of each echo:

$$\hat{m}_d(\Psi) = \sum_p \sqrt{h_p^2} e^{\lambda_p(\mathbf{d}_p \cdot \mathbf{r}_d - 1)}, \quad (12)$$

where $\mathbf{d}_p$ denotes the position of the p -th Gaussian peak on the unit sphere, h_p its amplitude coefficient (relative presence), and λ_p its angular concentration parameter (lobe sharpness). The vector $\mathbf{r}_d = [1, \Omega_d]$ denotes the direction associated with the d -th sample of the localization grid.

The model parameters are collected in the matrix $\Psi \in \mathbb{R}^{4 \times P}$, whose p -th column is defined as

$$\psi_p = [\Omega_p, h_p, \lambda_p]^\top.$$

The estimation problem therefore consists of finding the parameters Ψ that best describe the measured localization function. In particular, the estimated directions Ω_p provide the DoAs of the detected echoes, while the coefficients h_p provide a relative measure of their strengths.

The model can equivalently be written in matrix form as

$$\hat{\mathbf{m}}(\Psi) = \mathbf{M}\mathbf{h}, \quad (13)$$

where $\mathbf{h}$ contains the presence factors $\sqrt{h_p^2}$ and the entries of $M_{d,p}$ are given by $M_{d,p} = e^{\lambda_p(\mathbf{d}_p \cdot \mathbf{r}_d - 1)}$.

The use of $\sqrt{h_p^2}$ ensures that the amplitude coefficients remain nonnegative during the optimization.

The parameters Ω_p , h_p , and λ_p are estimated by minimizing the weighted least-squares objective

$$\mathcal{L}(\Psi) = \frac{1}{\mathcal{M}_{\text{tot}}} \sum_d \bar{m}_d [\bar{m}_d - \hat{m}_d(\Psi)]^2, \quad (14)$$

where $\mathcal{M}_{\text{tot}} = \sum_d \bar{m}_d$. This corresponds to a mean squared error (through a gradient descent on the linear system of equations described by Eq. (13)) weighted by the measured localization function $\bar{\mathbf{m}}$ itself. Consequently, errors in regions containing strong localization peaks contribute more strongly to the objective than errors in regions where no significant directional energy is present. This weighting is desirable because the objective is primarily concerned with accurately fitting the detected peaks, while avoiding over-fitting to background fluctuations in low-energy regions.

The optimization is performed using gradient descent with an adaptive step size determined through a secant-based estimate following the Barzilai–Borwein method [33].

D. Estimation of the Number of Incident Echoes

The localization procedure described above assumes that the number of peaks P is known when modelling the localization function. In practice, however, this quantity is unknown when analysing an SRIR frame in a blind manner. An additional procedure is therefore required to determine the appropriate number of Gaussian components to include in the model.

We propose an iterative estimation strategy based on an information criterion inspired by the Akaike information criterion (AIC) [34]:

$$\mathcal{C} = -\{\ln[\mathcal{L}(\Psi)] + P\}, \quad (15)$$

where $\ln(\cdot)$ denotes the natural logarithm. This criterion introduces a trade-off between the quality of the fit (absolute fitting error) and the complexity of the model. Increasing the number of Gaussian components generally reduces the reconstruction error, but additional components may simply model spatial fluctuations or noise rather than physically (and also perceptually) meaningful reflections. The use of an information criterion can, in principle, constrain the modelling process such that the selected number of radial Gaussian functions corresponds to the actual number of incident echoes. In practice, this relies on the assumption that physically meaningful reflections provide a sufficient reduction in fitting error to justify their inclusion, whereas additional components primarily capture spatial fluctuations or noise.

The criterion above was selected empirically based on its performance in the preliminary simulations presented in Section IV. A systematic comparison with alternative information criteria, such as the Bayesian information criterion (BIC) [35], is beyond the scope of this work. Nevertheless, the influence of the chosen criterion on the detection performance is briefly examined in Section IV-C.

The estimation procedure begins by fitting the localization function using a single Gaussian component, i.e. $P_0 = 1$, and evaluating the corresponding criterion value $\mathcal{C}_0$. The number of components is then increased iteratively according to $P_{i+1} = P_i + 1$ with the localization function re-estimated and the corresponding criterion $\mathcal{C}_{i+1}$ evaluated at each iteration. As additional Gaussian components are introduced, the criterion initially decreases as the fit improves, before increasing once the reduction in fitting error no longer justifies the increased model complexity. The iterative procedure is therefore terminated when $\mathcal{C}_{i+1} > \mathcal{C}_i$ and the model obtained at the previous iteration is selected as the final estimate of the number of incident echoes. The performance of this procedure is evaluated in Section IV-C.

IV. EVALUATION

This section evaluates the proposed method through both simulated and measured data. The objectives are twofold. First, the proposed Herglotz localization function, defined in Eq. (11), is compared with localization maps obtained using conventional SRP and MUSIC approaches. Second, the complete echo detection framework introduced in Sections III-C and III-D is evaluated in terms of its ability to estimate the directions, relative amplitudes, and number of incident echoes.

To ensure a fair comparison, the same peak detection procedure is applied to every localization map. Specifically, the radial Gaussian model of Eqs. (12) and (13), the gradient-based parameter estimation of Eq. (14), and the iterative AIC-based estimation of the number of peaks are used irrespective of the localization method. Since only a limited number of methods in the literature explicitly address peak detection and/or source number estimation, the comparisons presented below focus primarily on the quality of the underlying localization functions rather than on differences introduced by the peak extraction procedure.

A. Simulation Setup

The proposed method is first evaluated using simulated analysis frames representative of the early portion of an SRIR, i.e. $t \in [t_0, t_{\text{mix}}]$. Each frame consists of 256 samples (5.33 ms at a sampling frequency of 48 kHz) and contains a prescribed number P_{sim} of impulsive reflections.

Unless otherwise stated, four echoes are synthesized with incident directions $\mathbf{\Omega}_1 = [-120^\circ, 45.0^\circ]$, $\mathbf{\Omega}_2 = [22.5^\circ, 60.0^\circ]$, $\mathbf{\Omega}_3 = [90.0^\circ, -22.5^\circ]$, and $\mathbf{\Omega}_4 = [-11.25^\circ, -30^\circ]$. Their relative amplitudes are 0, -6, -2.5, and -12 dB with respect to the first echo.

Each echo is synthesized as temporal impulse (i.e. discrete Dirac distribution) bandpass-filtered between 125 Hz and 16 kHz. The lower cutoff avoids temporal aliasing caused by long low-frequency impulse responses within the finite analysis window. This bandwidth intentionally exceeds the maximally directive frequency band of the EM32 (≈ 800 –5200 Hz) in order to preserve the broadband characteristics of impulsive reflections.

An SMA measurement is simulated according to Eq. (1) using the geometry of the mh-acoustics Eigenmike. Unless otherwise stated, the simulations presented in this section use the EM32, while additional results obtained with the EM64 are presented later to evaluate the proposed method with a higher-order SMA. The simulated sound field is given by

$$u_{\text{sim}}(f, \mathbf{\Omega}_q) = \sum_{\ell=0}^{L_{\text{sim}}} b_{\ell}(f) \sum_{m=-\ell}^{\ell} Y_{\ell m}(\mathbf{\Omega}_q) \times \sum_{p=1}^{P_{\text{sim}}} u_p(f) Y_{\ell m}^*(\mathbf{\Omega}'_p), \quad (16)$$

where $\mathbf{\Omega}_q$ denotes the angular positions of the q -th microphone.

The simulated microphone signals are subsequently encoded into HOA coefficients using an SH transform followed by the application of the encoding filters $a_{\ell}(f)$ described in Sec. II-B [27]. The sound field is simulated with a truncation order of $L_{\text{sim}} = 20$, which is sufficiently high to avoid spatial aliasing below the temporal Nyquist frequency for the simulated Eigenmike geometry ($r_s = 4.2$ cm) at a sampling rate of 48 kHz. Consequently, the spatial aliasing observed in the simulated measurements is solely due to the finite spatial sampling of the array, and in particular to the truncation to $L = 4$ imposed by the $Q = 32$ microphone capsules of the Eigenmike.

B. Localization Function Estimation

Fig. 3 compares the final localization functions $\bar{\mathbf{m}}$ obtained from Eq. (11) using four different approaches to estimate the underlying frequency-dependent functions $\mathbf{m}(f_k)$ for the test signals described in Sec. IV-A. The proposed Herglotz-based localization function, obtained from Eq. (10), is shown in Fig. 3(a). For comparison, Fig. 3(b) shows a conventional SRP localization function based on a plane-wave decomposition (PWD) beamformer [29], Fig. 3(c) shows an SRP localization function obtained using a maximum weighted directivity index (maxWDI) beamformer [4], equivalent to the max- r_E beamformer described in [36], and Fig. 3(d) shows the MUSIC pseudospectrum [12]. The dimension of the MUSIC signal subspace was determined through eigenspectrum analysis following [13], resulting in an estimated number of $\hat{P}_{\text{MUSIC}} = 2$ sources.

The white stars indicate the true simulated echo DoAs. All spherical maps are visualized using the Equal Earth projection introduced by Šavrič et al. [37].

C. Echo Detection

Fig. 4 shows the radial Gaussian model $\hat{\mathbf{m}}(\Psi)$ obtained from Eqs. (12) and (13) fitted to the proposed Herglotz localization function $\bar{\mathbf{m}}$ [Eq. (11), Fig. 3(a)] for the simulated test frame described above. The white stars indicate the true simulated echo DoAs, and the spherical map is visualized using the Equal Earth projection [37].

Applying the proposed Gaussian fitting and model selection procedure to the Herglotz localization function correctly identifies the $P = 4$ simulated echoes in this example. The detection accuracy is evaluated using two complementary metrics. The first is the angular error between the synthesized DoA $\mathbf{\Omega}_p$ and the estimated direction $\tilde{\mathbf{\Omega}}_p$, obtained from the fitted unit vector $\tilde{\mathbf{d}}_p$ in Eq. (12). The second is the relative presence error, which compares the estimated presence factors $\hat{h}_p$ with the synthesized relative impulse levels after normalization with respect to the strongest reflection. For this example, the reference levels are 0, -6, -2.5, and -12 dB. The resulting errors for the Herglotz localization function are reported in the first row of Tab. I.

Table I summarizes the results obtained for the same simulated frame using the three reference localization

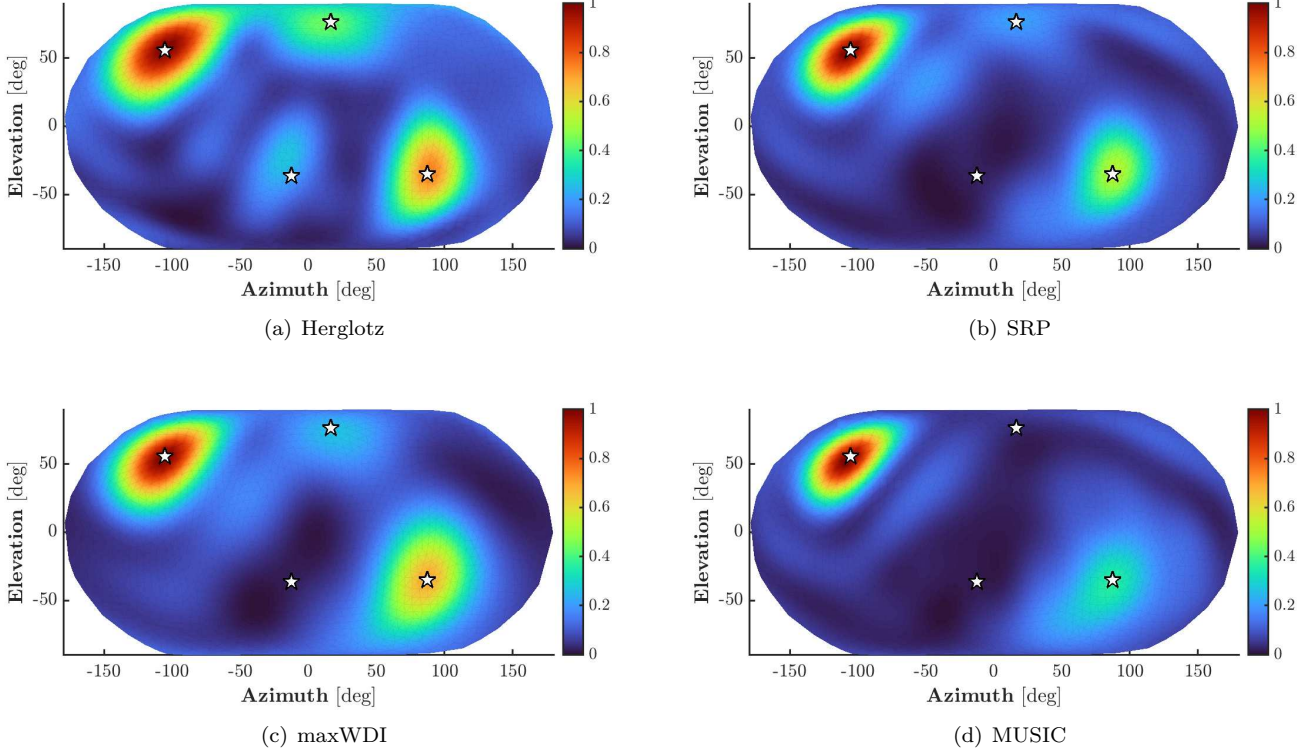


Fig. 3. Localization function $\bar{\mathbf{m}}$ calculated according to Eq. (11) for a simulated analysis frame containing $P_{\text{sim}} = 4$ echoes. The echoes are located at DOAs $\mathbf{\Omega}_1 = [-120^\circ, 45.0^\circ]$, $\mathbf{\Omega}_2 = [22.5^\circ, 60.0^\circ]$, $\mathbf{\Omega}_3 = [90.0^\circ, -22.5^\circ]$, and $\mathbf{\Omega}_4 = [-11.25^\circ, -30^\circ]$, respectively. The relative amplitudes of the second, third, and fourth echoes are set to -6 dB, -2.5 dB, and -12 dB with respect to the first echo. The spherical data are visualized using the Equal Earth projection [37].

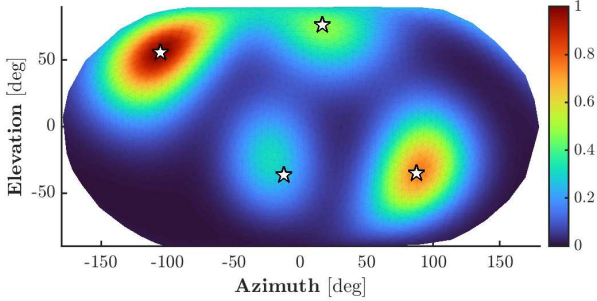


Fig. 4. Reconstructed localization function $\hat{\mathbf{m}}(\Psi)$ obtained from the radial Gaussian model of Eqs. (12) and (13) for the simulated four-echo frame described in Sec. IV-A. The $P_{\text{sim}} = 4$ echoes are placed at $\mathbf{\Omega}_1 = [-120^\circ, 45.0^\circ]$, $\mathbf{\Omega}_2 = [22.5^\circ, 60.0^\circ]$, $\mathbf{\Omega}_3 = [90.0^\circ, -22.5^\circ]$, and $\mathbf{\Omega}_4 = [-11.25^\circ, -30^\circ]$, respectively. The model parameters are estimated through gradient descent minimizing the objective function $\mathcal{L}(\Psi)$ given in Eq. (14). The white stars indicate the true simulated echo DOAs, and the spherical map is visualized using the Equal Earth projection [37].

functions (SRP, maxWDI, and MUSIC) along with the same radial Gaussian fitting and model selection procedure. The final column reports the detection rate for each method, defined as the ratio between the number of detected echoes and the number of synthesized echoes, expressed as a percentage.

The results indicate that, for the simulated SMA mea-

surement frame containing multiple discrete echoes described in Sec. IV-A, the Herglotz-based localization function provides the most accurate overall reconstruction among the methods considered. It achieves the lowest angular errors for three of the four reflections and consistently yields the smallest relative presence errors. The second reflection, synthesized at -6 dB relative to the strongest echo, represents the only case where another method (maxWDI) provides a lower angular error.

Furthermore, the proposed localization function is the only one for which the subsequent Gaussian fitting and model selection procedure correctly estimates the number of incident echoes in this example. This result should, however, be interpreted in relation to the information criterion (IC) used in the iterative procedure described in Sec. III-D, since the estimated model complexity depends on the chosen balance between fitting accuracy and parsimony. Inspection of the Herglotz localization map in Fig. 3(a) also suggests that the detection performance is ultimately limited by the dynamic range between incident echoes. As the level difference increases, weaker reflections may become indistinguishable from spatial noise or spatial aliasing artefacts generated by stronger components. Nevertheless, among the localization functions compared in Fig. 3, the Herglotz formulation provides the closest representation of the four simulated echo directions.

TABLE I
PEAK LOCALIZATION ACCURACY

Method	Angular Errors [deg]				Relative Presence Errors [dB]				Detection Rate [%]
Herglotz	0.773	2.23	2.93	15.4	0	0.0919	0.570	0.352	100
SRP	0.903	1.25	32.9	–	0	2.27	6.15	–	75
maxWDI	1.04	0.667	6.12	53.3	0	0.804	2.98	1.95	125
MUSIC	1.35	4.97	–	–	0	5.60	–	–	50

TABLE II
PEAK LOCALIZATION ACCURACY, EM64 SIMULATION

Method	Angular Errors [deg]						Relative Presence Errors [dB]						Detection Rate [%]
Herglotz	0.14	0.729	0.427	1.1624	0.9169	–	0	0.002	0.405	0.92	0.426	–	83
SRP	0.403	1.573	3.024	–	4.475	–	0	1.609	2.027	–	1.021	–	67
maxWDI	3.328	3.821	3.571	10.6826	8.916	–	0	2.143	3.8631	0.319	3.165	–	150
MUSIC	1.851	1.622	3.636	–	–	–	0	2.81	0.058	–	–	–	50

D. Validation with a Higher-Order SMA

To assess the sensitivity of the proposed method to the spatial order of the measurement array, the simulation described in Sec. IV-A was repeated using the geometry of the mh-acoustics EM64 ($Q = 64$ microphone capsules, truncation order $L = 6$) rather than the EM32. The number of synthesized echoes was increased to $P_{\text{sim}} = 6$, placed at $\Omega_1 = [-120^\circ, 45.0^\circ]$, $\Omega_2 = [22.5^\circ, 60.0^\circ]$, $\Omega_3 = [90.0^\circ, -22.5^\circ]$, $\Omega_4 = [-87.0^\circ, -60.0^\circ]$, $\Omega_5 = [-55.0^\circ, 12.0^\circ]$, and $\Omega_6 = [0^\circ, 0^\circ]$, with relative amplitudes of 0, -2.5 , -4.5 , -7 , -10.5 , and -16.5 dB with respect to the first echo.

Fig. A.1 compares the localization functions obtained with the four methods considered in Sec. IV-A for this six-echo frame. As with the EM32 case, the Herglotz localization function yields sharper, better-separated peaks at the true echo directions (white stars) than the SRP, maxWDI, and MUSIC alternatives, which increasingly merge or misplace the weaker reflections as the number of concurrent echoes grows.

Table II reports the corresponding peak localization accuracy. Consistent with the EM32 results of Sec. IV-C, the Herglotz-based localization function achieves the lowest angular error for four of the five reflections it detects, together with the smallest relative presence errors overall, and the highest detection rate (83%) among the four methods. The MUSIC pseudospectrum, whose signal-subspace dimension is again constrained by eigenspectrum analysis, fails to resolve the weaker reflections and yields the lowest detection rate (50%), while SRP and maxWDI fall between these extremes, with maxWDI over-segmenting the localization map into spurious peaks (detection rate above 100%).

E. Large-Scale Statistical Simulation

To evaluate the proposed method beyond individual example frames, a large-scale statistical simulation was conducted. Individual analysis frames containing between

one and twelve reflections were simulated as described in Sec. IV-A, using the EM64 geometry. For each number of reflections $P_{\text{sim}} \in \{1, \dots, 12\}$, 250 independent scenarios were generated with randomly drawn incident directions, subject to a minimum angular separation of 35° between echoes to avoid ambiguity from overlapping reflections. All reflections within a given frame were simulated with equal amplitude, isolating the influence of the number of concurrent echoes from that of their relative levels.

Fig. A.2 illustrates the localization functions obtained with the four methods for a representative scenario containing $P_{\text{sim}} = 12$ randomly placed, equal-level echoes. As in the lower-order cases, the Herglotz localization function preserves twelve visually distinguishable peaks, whereas the reference methods exhibit greater smearing and spurious structure at this echo density.

Fig. 5 summarizes, for each simulated number of reflections, the distribution of the number of echoes detected by the iterative Gaussian model-selection algorithm of Sec. III-D, applied to each of the four localization functions across the 250 configurations. Fig. 6 shows the corresponding medians of the mean angular error and relative presence error, computed over the correctly matched, detected reflections in each frame. Notably, the Herglotz-based localization function detects substantially more reflections than SRP and MUSIC, particularly as the number of reflections increases. The resulting median errors at these higher reflection counts are therefore especially indicative of the method’s ability to localize a larger number of concurrent reflections while maintaining accurate parameter estimates.

F. Measurement Validation

The proposed method was further validated using measured SRIRs acquired in the “Espace de projection” (ESPRO), IRCAM’s variable-acoustics concert hall [38]. ESPRO is a shoebox-shaped room whose electroacoustic architecture allows continuous adjustment of the room’s reverberant characteristics [39], [40]. The measurements

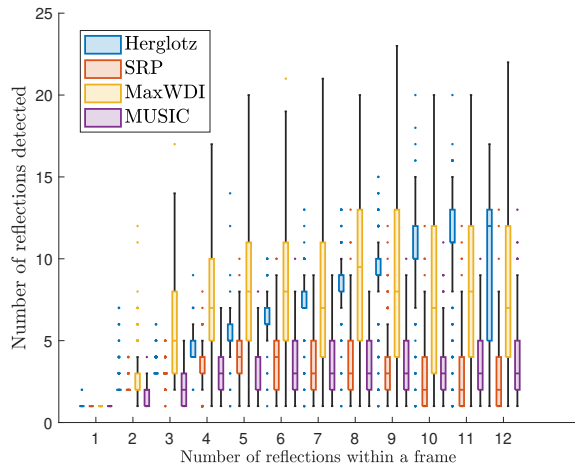


Fig. 5. Number of echoes detected by the iterative Gaussian algorithm of Sec. III-D, evaluated across 250 DoA configurations for each number of simulated reflections, using the four localization map methods.

analyzed here were collected as part of a larger measurement campaign in this space. The loudspeaker setup used during the campaign comprised a 75-loudspeaker HOA dome (Amadeus PMX-5 5-inch co-axial speakers), a nine-unit overhead speaker ring (d&b E12 12-inch speakers), and four additional loudspeakers (d&b E12 12-inch speakers) distributed on the floor. The microphone setup comprised ten ceiling-facing cardioid microphones (Neumann KM184), a binaural dummy head (Neumann KU100), and SMAs (mh-acoustics EM32 and EM64).

Two complementary evaluations were carried out on the measured EM64 SRIRs: a short-frame comparison focused on direct sound and early reflection localization accuracy, and a longer-frame comparison assessing the detection of a denser reflection sequence.

1) *Direct Sound and First Reflection*: Fig. A.4 shows the localization functions obtained with the four methods for a 10 ms analysis frame containing the direct sound and the first reflection of a measured EM64 SRIR. Table III reports the corresponding angular and relative presence errors, where the reference (“true”) DoAs were obtained from an Image Source Method (ISM) simulation using the known loudspeaker and microphone positions.

All four methods correctly detect and localize the DoA of the direct sound and the first reflection, confirming that, for early, well-separated arrivals, the choice of localization function has limited impact on detection reliability. MUSIC achieves the lowest angular error for the direct sound, while the Herglotz method achieves the lowest angular error for the first reflection and the lowest relative presence error for both arrivals.

2) *Denser Reflection Sequence*: To assess performance under more realistic, denser reflection conditions, the analysis was extended to a 42 ms frame excluding the direct sound and first two reflections but incorporating the following 13 reflections, whose reference DoAs were

computed using second-order ISM. Fig. A.3 shows the reconstructed localization maps obtained after applying the Gaussian detection algorithm of Sec. III-D to each of the four localization functions, with white stars marking the ISM-predicted reflection DoAs. Table IV summarizes the results.

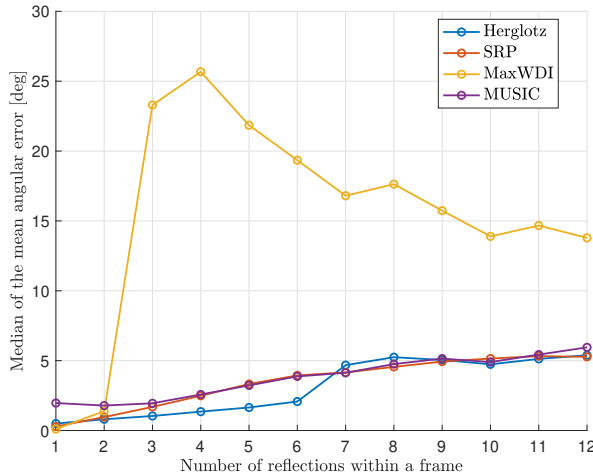
In this denser, more realistic reflection sequence, the Herglotz localization function substantially outperforms the reference methods, achieving a detection rate of 70% against 31–62% for SRP, maxWDI, and MUSIC. This measured result is consistent with the trends observed in the large-scale simulation of Sec. IV-E: as the number of concurrent, closely-spaced reflections within a single analysis frame grows, the proposed Herglotz formulation increasingly outperforms conventional SRP and subspace-based localization functions.

Taken together, the results of Secs. IV-D–IV-F indicate that, while all four localization methods perform comparably for isolated, well-separated arrivals such as the direct sound and first reflection, the proposed Herglotz localization function offers a marked and consistent advantage as the echo density increases, in both higher-order simulated conditions and real SRIR measurements acquired in a variable-acoustics concert hall. This advantage is observed across all three complementary metrics considered throughout this section, namely angular error, relative presence error, and detection rate, particularly once at least four echoes are present within the same analysis frame.

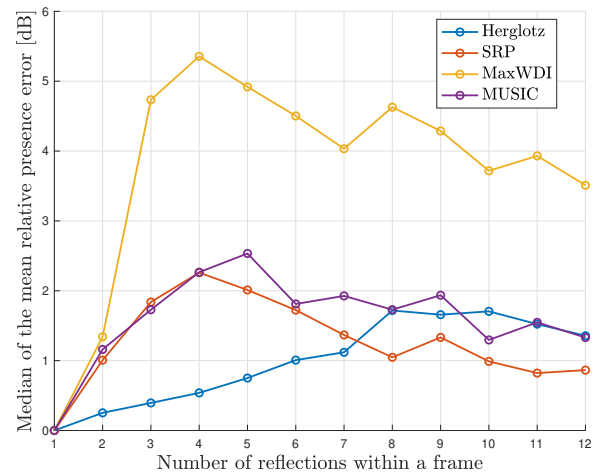
V. CONCLUSION

This paper introduced a Herglotz wavefunction-based framework for localizing multiple early reflections within a single analysis frame of an SRIR measured by an SMA. By representing the measured sound field as a superposition of Herglotz wavefunctions, the proposed formulation naturally leads to a localization function whose peaks correspond to the directions of arrival of the incident echoes. A regularized inversion in the spherical harmonic domain was developed to estimate this localization function, followed by a radial Gaussian fitting procedure and an iterative information criterion-based strategy to jointly estimate both the directions of arrival and the number of incident echoes.

Simulations and measurements demonstrated that the proposed localization function provides a more faithful representation of multiple simultaneous reflections than conventional SRP- and MUSIC-based localization maps. Combined with the proposed Gaussian fitting procedure, the method achieved more accurate direction estimates, lower relative amplitude (presence) errors, and more reliable estimation of the number of incident echoes, particularly in analysis frames containing several closely spaced reflections. These results suggest that the Herglotz formulation is well suited to the analysis of the early part of SRIRs, where multiple broadband impulsive arrivals coexist within the same time frame.



(a) Angular Error



(b) Relative Presence Error

Fig. 6. Medians of the mean angular error and relative presence error for detected reflections within a frame, evaluated across 250 DoA configurations for each number of simulated reflections, using the four localization map methods.

TABLE III
PEAK LOCALIZATION ACCURACY MEASUREMENT

Method	Angular Errors [deg]		Relative Presence Errors [dB]		Detection Rate [%]
Herglotz	1.5972	1.0169	0	0.719	100
SRP	1.9586	3.6250	0	3.436	100
maxWDI	2.4873	6.2975	0	3.614	100
MUSIC	0.7771	2.8962	0	5.752	100

TABLE IV
ECHO DETECTION ACCURACY, ESPRO MEASUREMENT
(13-REFLECTION FRAME)

Method	Detection Rate [%]
Herglotz	70
SRP	31
maxWDI	62
MUSIC	38

The present work focuses on the localization of reflections within individual analysis frames as a first step toward a complete spatio-temporal analysis of early reflections. Future work will investigate frame-to-frame echo tracking and the construction of complete early-reflection cartographies, along with a more extensive evaluation using both simulated and measured SRIRs. Further studies should also examine the influence of the selected information criterion, the dynamic range between reflections, and the maximum spherical harmonic order supported by the SMA on localization and detection performance.

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APPENDIX A WIDE COMPARISON PLOTS

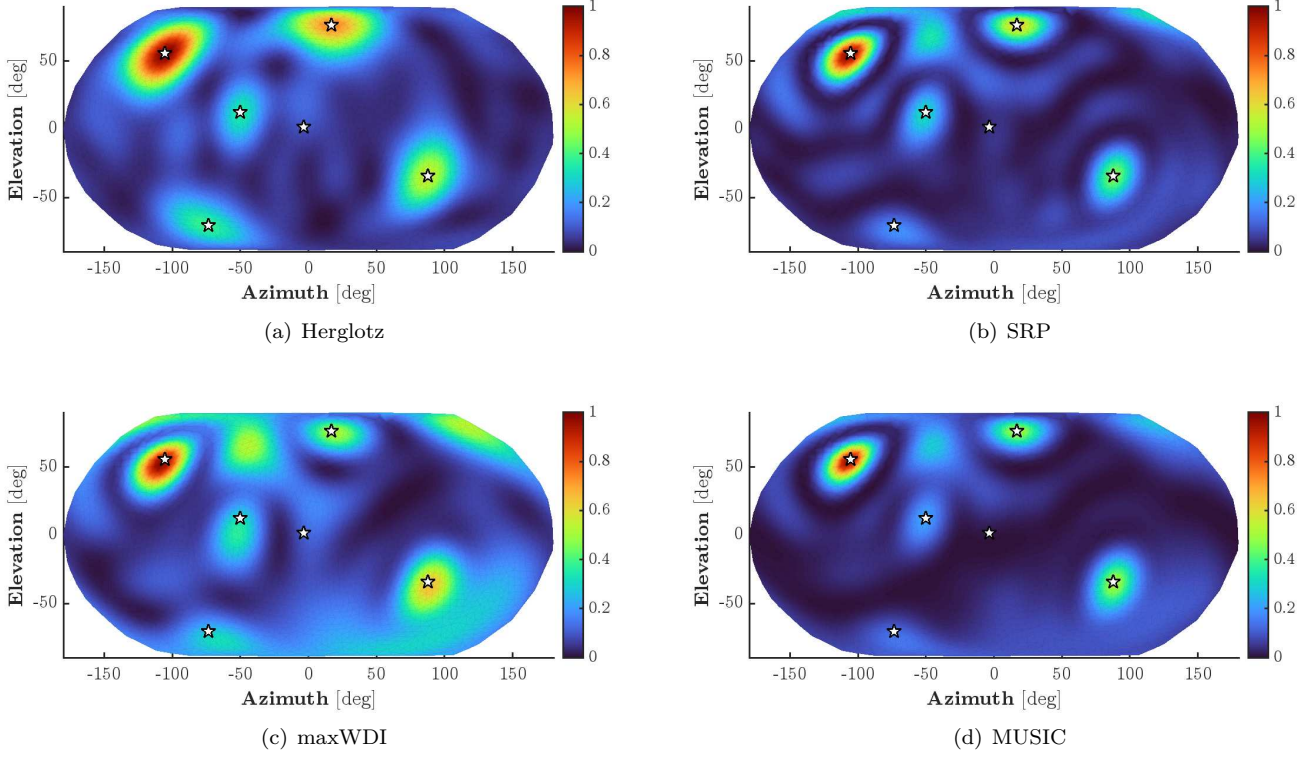


Fig. A.1. Localization function $\overline{\mathbf{m}}$, calculated following Eq. (11), for a simulated frame containing $P_{\text{sim}} = 6$ echoes placed at $\Omega_1 = [-120^\circ, 45.0^\circ]$, $\Omega_2 = [22.5^\circ, 60.0^\circ]$, $\Omega_3 = [90.0^\circ, -22.5^\circ]$, and $\Omega_4 = [-87.0^\circ, -60.0^\circ]$, $\Omega_5 = [-55.0^\circ, 12.0^\circ]$, $\Omega_6 = [0^\circ, 0^\circ]$ respectively. The second through sixth echoes are synthesized at -2.5 dB, -4.5 dB, -7 dB, -10.5 dB and -16.5 dB relative to the first. White stars indicate the true simulated DoAs. The sphere is shown using the Equal Earth projection [37].

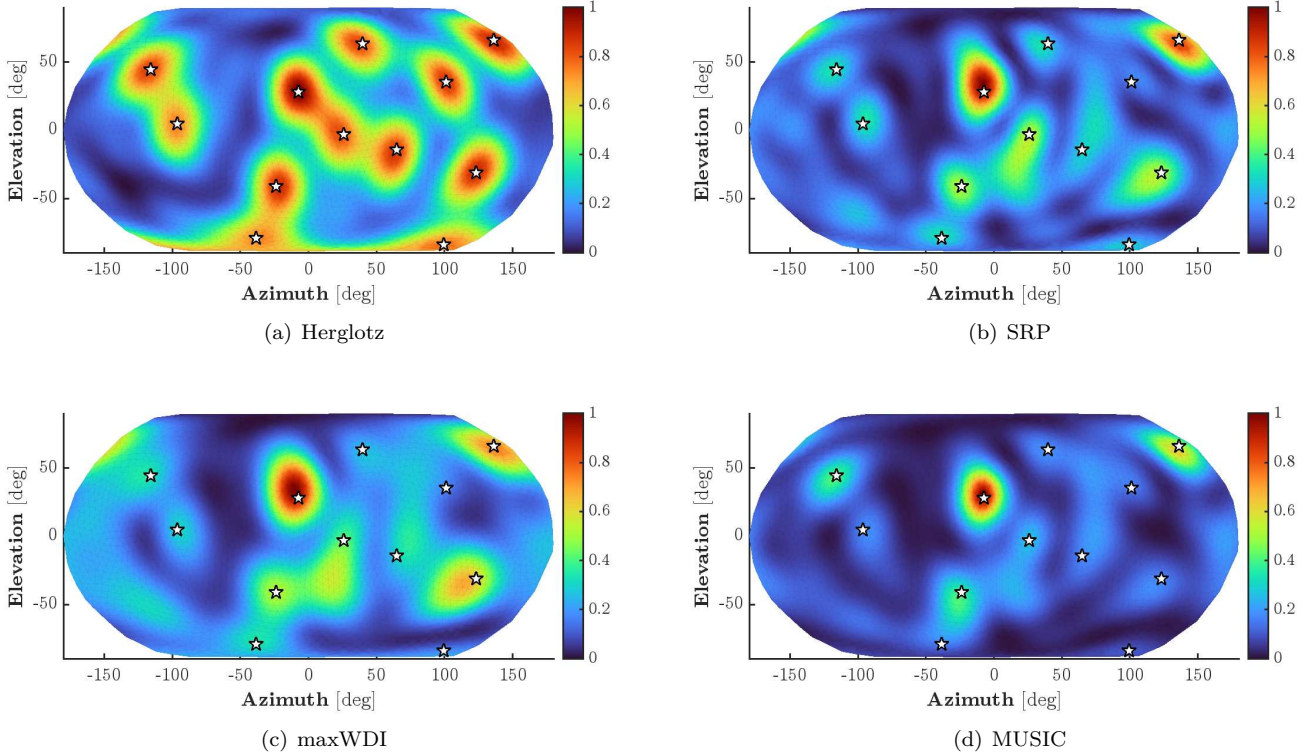


Fig. A.2. Localization function $\overline{\mathbf{m}}$, calculated following Eq. (11), for a simulated frame containing $P_{\text{sim}} = 12$ echoes randomly placed with a minimum angular separation of 35° . All twelve impulses share the same gain. The sphere is shown using the Equal Earth projection [37].

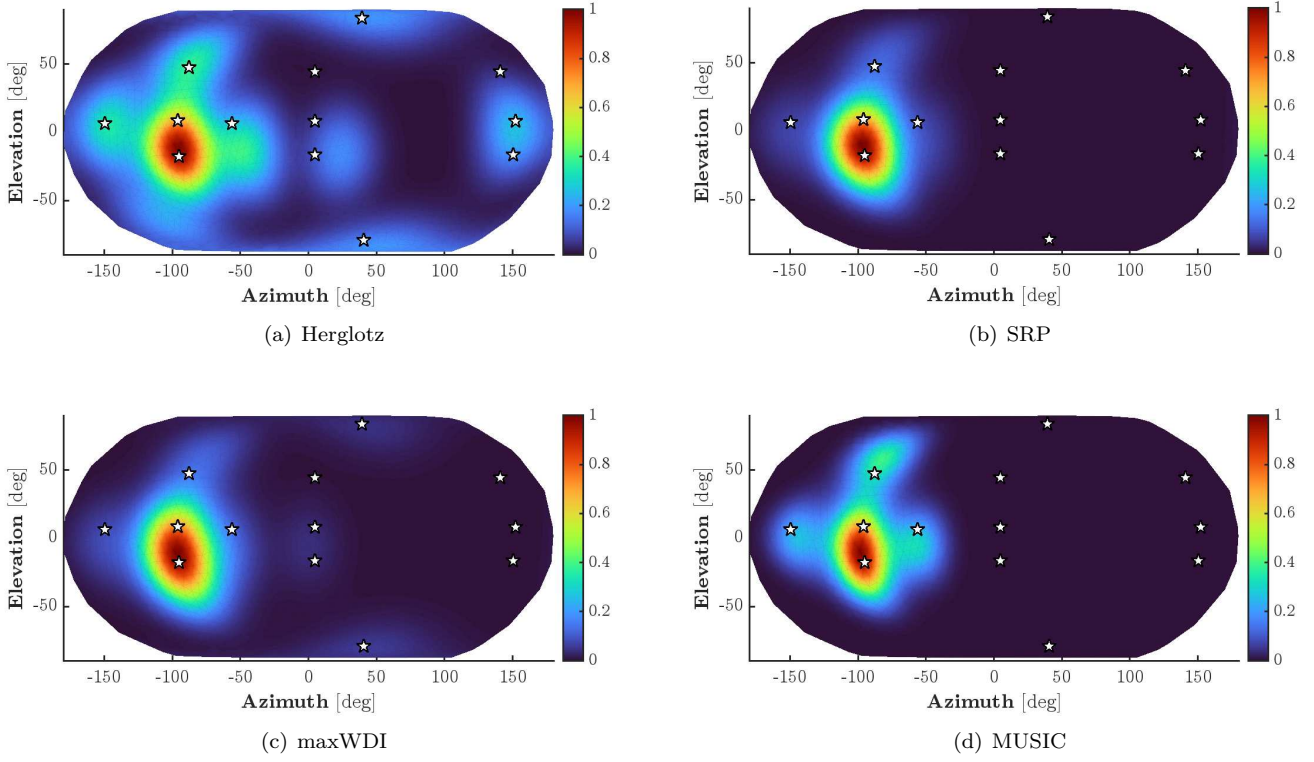


Fig. A.3. Reconstructed localization function $\hat{\mathbf{m}}(\Psi)$ obtained from the radial Gaussian model of Eqs. (12) and (13) applied to the four localization functions, computed from a measured EM64 SRIR acquired in ESPRO. The visualization corresponds to a 42 ms frame excluding the direct sound and the first two reflections, and incorporating the following 13 reflections predicted by second-order ISM. White stars indicate the ISM-predicted reflection DoAs. The sphere is shown using the Equal Earth projection [37].

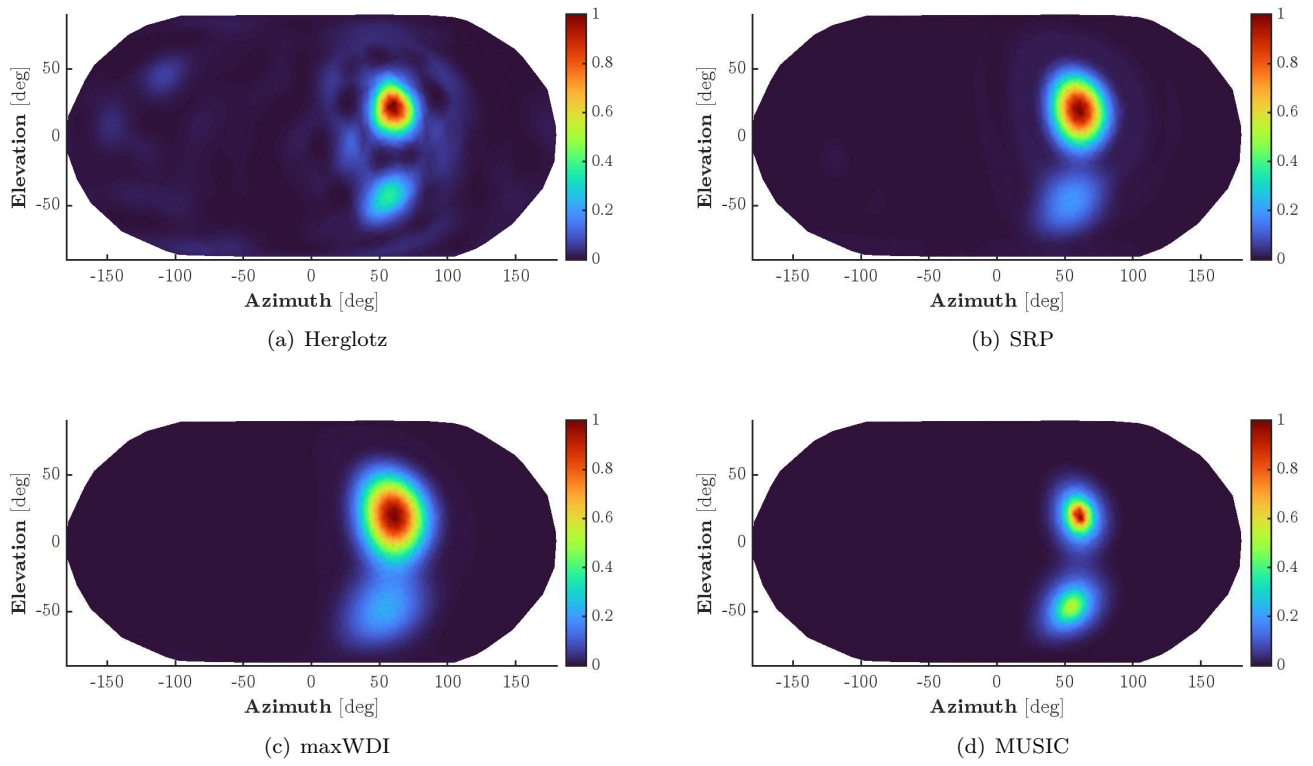


Fig. A.4. Localization function $\overline{\mathbf{m}}$, calculated following Eq. (11), from a measured EM64 SRIR acquired in ESPRO. The visualization corresponds to a 10 ms frame that includes both the direct sound and the first reflection. The sphere is shown using the Equal Earth projection [37].